

Autonomous Choke Control Using a Physics-Based Digital Twin and Intelligent Control Framework

- **Problem Statement ID** – PS3
- **Problem Statement Title** – Autonomous Production Choke Controller for a Single Naturally Flowing Oil Well
- **Theme** – Smart Automation / Industrial IoT
- **PS Category** – Software
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IDEA TITLE

Problem :

- Choke valves are one of the most critical components in an oil production system. Their operation directly affects oil production, wellhead pressure, equipment safety, and reservoir health.
- Conventional choke control is either manually operated or relies on fixed control strategies that cannot adapt to changing reservoir conditions.

Key Features :

- Physics-Based Digital Twin of the complete production system
- Five Intelligent Choke Control Strategies
- Real-Time Controller Benchmark Framework
- Interactive Monitoring Dashboard
- Safe Evaluation without Field Testing

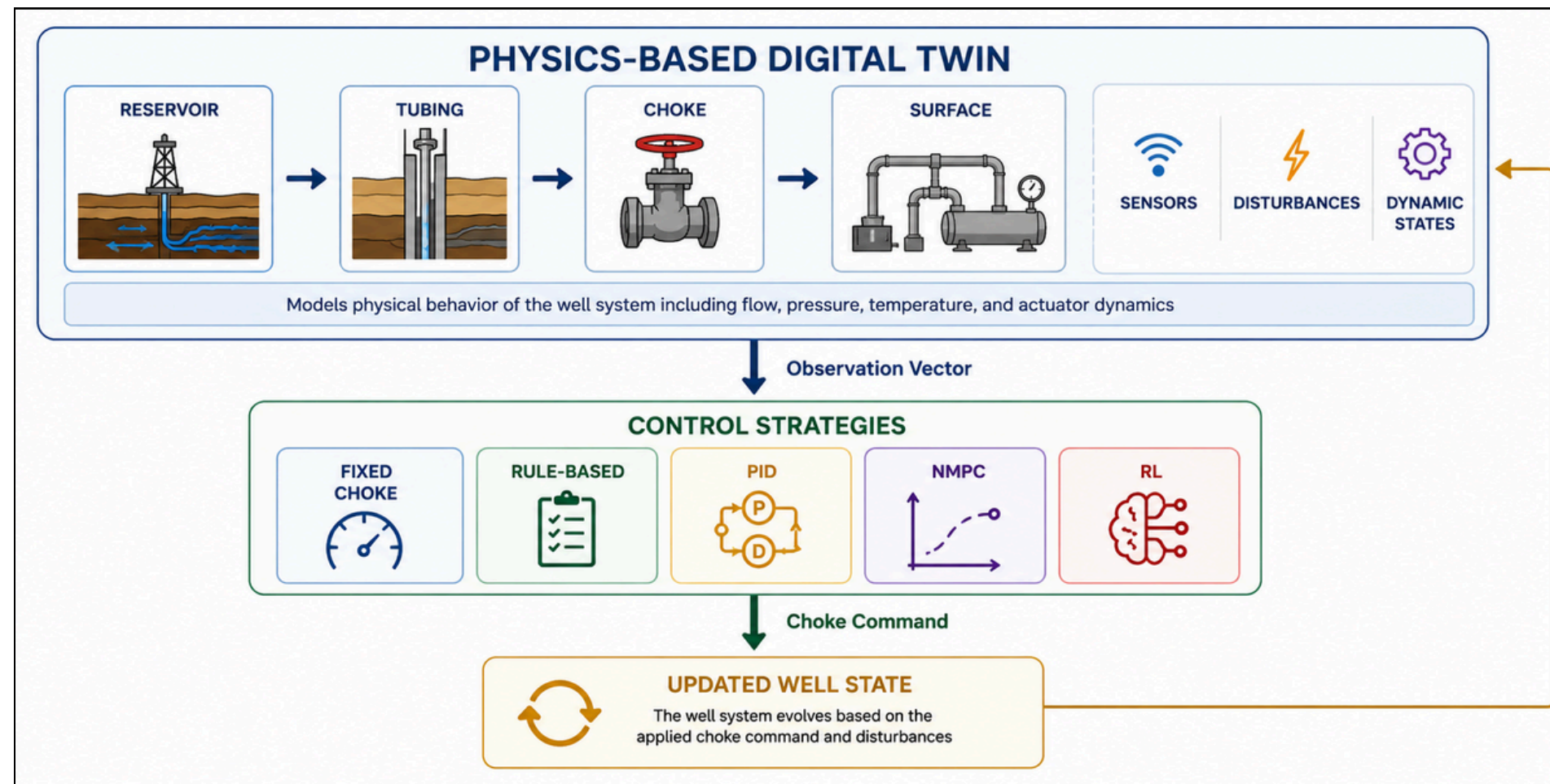
Proposed Solution :

- We propose an Autonomous Choke Control Platform that intelligently determines the optimal choke opening using a Physics-Based Digital Twin and multiple intelligent control algorithms.
- Instead of developing only one controller, our platform provides a common environment to design, validate, compare and benchmark different autonomous choke control strategies before real-world deployment.

Innovation & Uniqueness :

Unlike existing solutions that focus on a single controller, our platform combines realistic well simulation with multiple intelligent control techniques, enabling fair comparison and selection of the most suitable controller for different operating conditions.

TECHNICAL APPROACH



- **Intelligent Choke Controllers**

Every controller interacts with the same Digital Twin.

- Fixed Choke
- Rule-Based Controller
- PID Controller
- Nonlinear Model Predictive Controller
- Reinforcement Learning Controller

- **Benchmark & Validation**

- Controllers are evaluated using identical operating conditions.
- Performance Metrics
 - Oil Production
 - Wellhead Pressure
 - Tracking Accuracy
 - Valve Wear
 - Solver Stability
 - Computational Latency

- **Technologies Used**

- Python
- NumPy
- SciPy
- Gymnasium
- Matplotlib



FEASIBILITY AND VIABILITY

Controller Framework

To identify the most effective autonomous choke control strategy, five controllers were implemented and evaluated on the validated Digital Twin.

Controller	Purpose
Fixed Choke	Baseline reference
Rule-Based	Expert heuristic control
PID	Classical feedback control
NMPC	Predictive optimization-based control
Reinforcement Learning	Adaptive intelligent control

Benchmark Scenarios

- All controllers were tested under identical operating conditions:
 - Production Setpoint Changes
 - Separator Pressure Disturbances
 - Water Cut Variations
 - Reservoir Pressure Decline
- This ensured a fair and consistent comparison of controller performance.

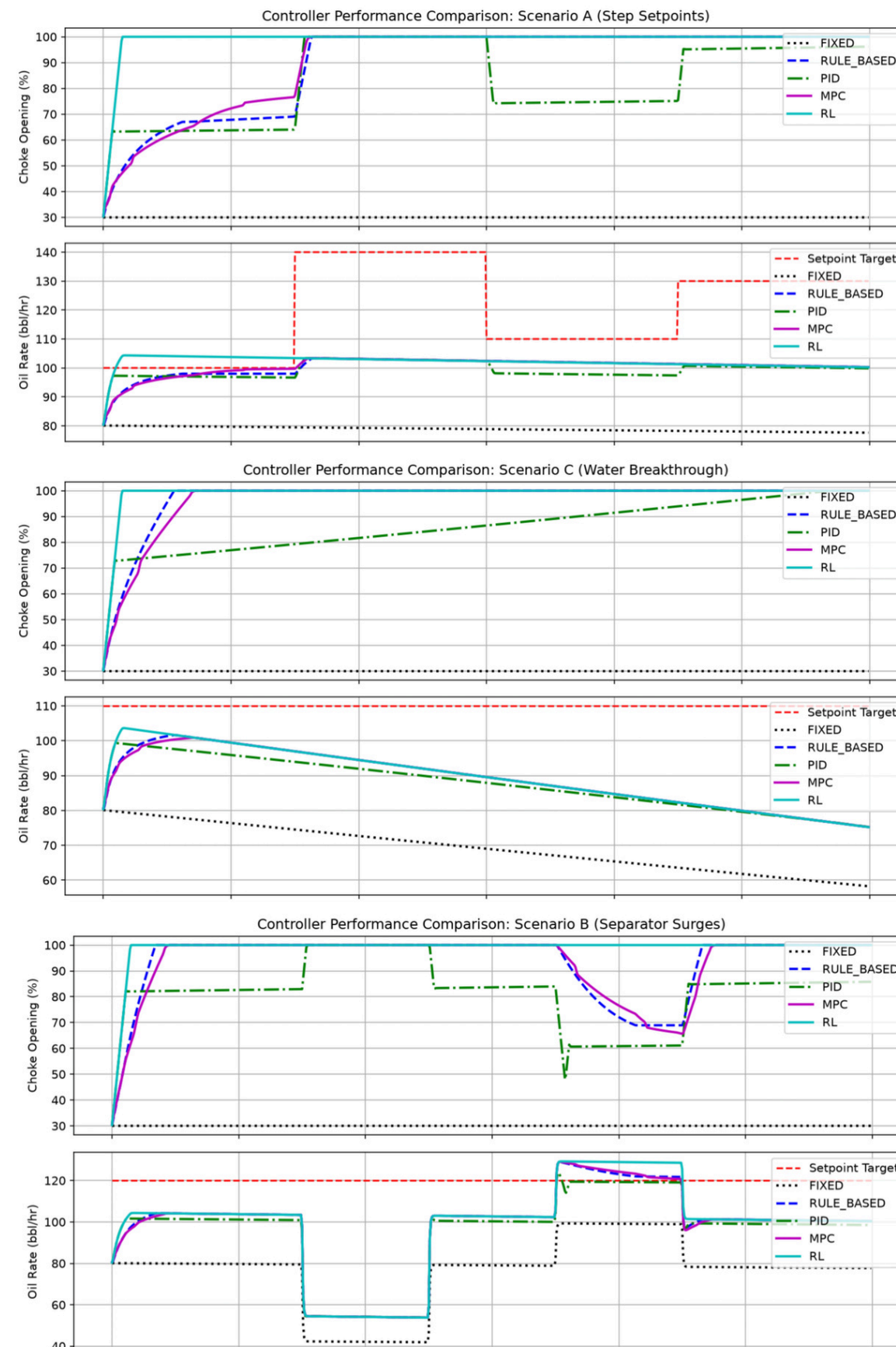
Key Outcomes

- Successfully benchmarked five autonomous choke control strategies.
- Demonstrated safe operation under dynamic well conditions.
- Identified the strengths and trade-offs of each control approach.
- Established a reusable framework for future controller development and evaluation.

STATISTICAL SUMMARY TABLE (MEAN ± STD DEV across N = 5 Trials)			
Controller	Cum Oil Production (bbl)	Tracking Error IAE (bbl·h)	Choke Wear (%)
FIXED	926.4 ± 93.1 bbl	511.6 ± 93.1 bbl·h	0.0 ± 0.0%
RULE_BASED	1201.8 ± 120.9 bbl	236.2 ± 120.9 bbl·h	69.7 ± 0.0%
PID	1176.8 ± 102.7 bbl	261.2 ± 102.7 bbl·h	645.8 ± 274.6%
MPC	1201.7 ± 120.9 bbl	236.3 ± 120.9 bbl·h	68.7 ± 0.0%
RL	1204.3 ± 122.1 bbl	233.7 ± 122.1 bbl·h	69.3 ± 0.0%
Lower is better (Tracking Error, Choke Wear) Higher is better (Cum Oil Production) High wear (Higher is worse)			

HYPOTHESIS TESTING (Two-Tailed Student's t-test p-values)			
Comparison	t-statistic	p-value	Conclusion (α = 0.05)
RL vs NMPC	0.030	0.9768	p >= 0.05 Not Statistically Significant
RL vs Rule-Based	0.029	0.9774	p >= 0.05 Not Statistically Significant
NMPC vs Rule	-0.001	0.9993	p >= 0.05 Not Statistically Significant
Interpretation: All p-values are greater than 0.05, which indicates that there is no statistically significant difference in cumulative oil production between RL, NMPC, and Rule-Based controllers at the 5% significance level.			

ARTIFACTS



Dashborad

Implemented Features

- ✓ Real-Time Dashboard
- ✓ Three Benchmark Scenarios
- ✓ Monte Carlo Analysis
- ✓ Statistical Evaluation
- ✓ Interactive Controller Comparison
- Complete benchmark plots, Monte Carlo analysis, and statistical results are available in the [project repository](#).

RESEARCH AND REFERENCES

Our solution is based on established research in petroleum production engineering, choke valve hydraulics, digital twins, optimal control, and reinforcement learning.

Topic	Reference
Petroleum Production Engineering	Economides, M. J., <i>Petroleum Production Systems</i>
Choke Valve Flow Modelling	API RP 14E – Recommended Practice for Design and Installation of Offshore Production Platform Piping Systems
Digital Twin	Grieves, M. (2014), <i>Digital Twin: Manufacturing Excellence through Virtual Factory Replication</i>
Model Predictive Control	Rawlings, J. B., <i>Model Predictive Control: Theory and Design</i>
Reinforcement Learning	Sutton, R. S. & Barto, A. G., <i>Reinforcement Learning: An Introduction (2nd Edition)</i>

Project Repository
Source Code, Benchmark Results, Plots & Statistical Analysis
 <https://github.com/atharvadhk/autonomous-choke-control>